



13.H1.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about each learning target.

1. I can determine the k -value of a transformation.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



2. I can describe a transformation using words or function notation.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



Unit 13, Lesson 8: Identifying and Interpreting Parts of an Expression or Equation – Part 1



Benchmark

MA.912.AR.1.1 Identify and interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context, including viewing one or more of its parts as a single entity.



Learning Targets

- ☐ I can identify parts of an expression including factors, terms, constants, coefficients, and variables.
- ☐ I can interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context.



13.8.1 Warm-Up: Florida Scrub-Jay

The Florida scrub-jay is a blue and gray bird about the size of a blue jay. The Florida scrub-jay is not only native to Florida, but it only lives in Florida. Scrub-jays inhabit the ancient sandy ridges that run down the middle of the state, old sand dunes along Florida's coasts, and sandy deposits along the rivers in the interior of the state. In the absence of lightning-ignited, low-intensity fires that used to be common in Florida, the scrub-jay's natural habitat has become too dense and tall to support them. Before Florida was settled in 1800, the bird's population is estimated to have been 54,000. It's estimated that by 1890 the population was 27,600. More recently, the estimates of the Florida scrub-jay population were 4,000 in 1993 and 3,000 in 2010.



1. What is the scrub-jay's population decline from 1800 to 1890 as a percentage?
2. What is the rate of decay from 1993 to 2010?
3. During which time period did the most devastating decrease in the Florida scrub-jay population occur?



13.8.2 Guided Instruction: Parts of an Equation or Expression

1. Jacquie is hiking a section of the Florida Trail, starting at the trailhead located at historic Fort Pickens in Gulf Islands National Seashore. Jacquie packs trail mix and beef jerky for snacks. The trail mix provides 3.25 calories per gram and the beef jerky provides 2.7 calories per gram. She would like to only consume 158 calories.

- a. $3.25m + 2.7b$ is a(n)

- equation.
- expression.
- function.

- b. The values 3.25 and 2.7 are

- constants.
- coefficients.
- terms.

- c. The monomials $2.7b$ and $3.25m$ are

- constants.
- coefficients.
- terms.

- d. $3.25m + 2.7b = 158$ is a(n)

- equation.
- expression.
- function.

- e. Determine the amount of trail mix Jacquie can eat if she eats 20 grams of beef jerky. Show your work.

2. U.S. food availability is determined by taking how much of a certain food is produced and subtracting exports to other countries. In 1970, the amount of cheese available for consumption was 17.22 pounds per person. For each year after 1970, the amount of cheese that was available increased by 0.48 pounds per person.

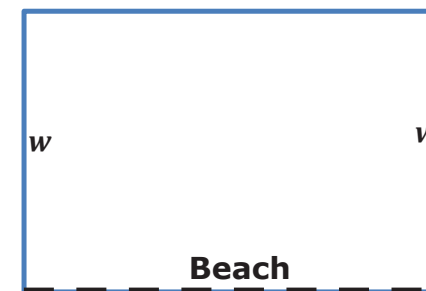
- a. This situation represents
relationship.

- ☐ a linear
- ☐ a quadratic
- ☐ an exponential

- b. Complete the table.

Quantity	Description
0.48	☐ initial value ☐ rate of change ☐ informational only
17.22	☐ initial value ☐ rate of change ☐ informational only
1970	☐ initial value ☐ rate of change ☐ informational only

3. A lifeguard uses 2,000 feet of rope to create a safe rectangular-shaped swimming area in the ocean. The diagram shows the width, w , of the swimming area and the location of the beach. The rope is not used alongside the beach.



Explain what each quantity represents in this context.

Quantity	Explanation
$2000 - 2w$	
$w(2000 - 2w)$	

4. Microplastics are buoyant plastic materials smaller than 0.5 centimeters in diameter. The amount of these plastics in the ocean since 1994 can be modeled by the equation $y = 100,100(1 + 0.073)^x$, where y is the number of tons of plastics in the ocean and x the number of years since 1994.

- a. Discuss with your partner why the values 0.5 and 1994 are not in the equation $y = 100,100(1 + 0.073)^x$. Complete the table.

Value	Explanation
0.5	
1994	

- b. Ask a classmate for their explanation. Record their explanation. *You are the only person who should write in the second column of the table.* Have the person initial next to your summary stating that your summary is correct.

Value	My Summary of Their Explanation	Initials
0.5		
1994		

- c. Complete the sentences.

- The equation $y = 100,100(1 + 0.073)^x$ represents

- ☐ exponential decay.
- ☐ exponential growth.

- The rate of ☐ decay ☐ growth is _____%.

- The value 100,100 is the ☐ amount of plastics ☐ tons of microplastics found in the ocean in 1994.



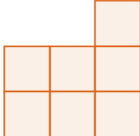
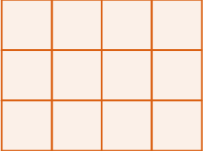
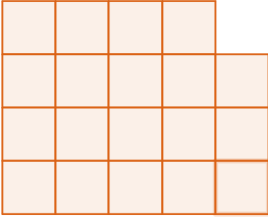
13.8.3 Guided Practice: Parts of an Equation or Expression

- An orange grove manager uses two different-sized buckets to load oranges into a truck. The first bucket can transport x pounds of oranges, and the second bucket can transport y pounds of oranges. The first bucket is filled up G times and the second is filled up H times.

Explain what each quantity means.

Quantity	Explanation
$G + H$	
$x + y$	
$Gx + Hy$	

- A pattern is shown. The number of tiles in each step is represented by the expression $(n + 1)^2 + 3$, where n is the step.

Step 1	Step 2	Step 3
		

Circle the part of each diagram that is represented by the expression $(n + 1)^2$.



13.8.4 Your Turn!

The function $C(x) = 0.48x + 17.22$ models the amount of cheese, C , in pounds, available for consumption for each year since 1970.

1. What does the slope of the function, 0.48, represent in this context?
2. Which value is a constant?
3. What does the constant represent?
4. What does the variable x represent?



13.8.5 Wrap-Up: Reflection

1. Complete the table.

Today I deserve  for ...	
Today I  by doing ...	
Today I was  by ...	